VIDEO GAME CREATION

Lesson 11 - Core Mechanics V: Wave After Wave

Semester 1, Week 11 - Level 2 Complete Milestone

- **Objectives:** Nest a second decision inside an existing conditional (is this the last wave?); reuse the spawn_enemies routine from a third caller, the game itself, not just launch and player input; keep new shared state (wave) on the GameManager and reset it on restart; explain what completing the Level 2 Complete milestone means for the capstone.
- **IGDA Tie-In:** Programming - state machines, code reuse/refactoring, game balance and pacing (intermediate)
- **Featured Engine (lecture demo):** Unity (C#)
- **Materials:** Demo laptop with the four active engines verified (see Teacher's Checklist); Student Handouts for Scratch, Godot, GameMaker (script and Visual) and Unity; students' own Lesson 10 projects (already saved) - no new assets needed this week

Agenda

0:00-0:05	Welcome back, Lesson 10 recap (game state, the Stalemate)
0:05-0:20	One More Question: Is This the Last Wave?
0:20-0:40	The Same Spawn Code, Called a Third Way (live demo in Unity)
0:40-0:50	Milestone Moment: What "Level 2 Complete" Actually Means
0:50-0:55	Introduce lab exercise, hand out engine-specific handouts
0:55-1:50	LAB: Build multiple waves, reaching the Level 2 Complete milestone
1:50-2:00	Exit check / quick share-out - milestone celebration

Levels for this lesson's exercise

- Level 1 (Minimal): clearing a wave spawns a fresh one, up to 3 total; clearing wave 3 triggers the real win; wave number shown on screen.
- Level 2 (Complete): wave transitions are seamless and correct every time - no double-spawns, no skipped waves, restart always returns to wave 1.
- Level 3 (Refined): fix the Stalemate (left in on purpose from Lesson 10) so it works with waves; add a brief wave-transition announcement; and/or tighten the reusable spawn structure.

See: Lesson 11 Teacher's Script, Lesson 11 Prep Checklist, and the per-engine Student Handouts. This is a milestone lesson - plan extra time for the Milestone Moment and end-of-lesson celebration, same as Lesson 6.